SAFETY DATA SHEET

SECTION 1 : CHEMICAL & COMPANY IDENTIFICATION

PRODUCT NAME	:	Amoil NMP
Chemical Family	:	Heterocyclic, Amides
SUPPLIER	:	Amoil (Malaysia) Sdn.Bhd.
ADDRESS	:	14, Jalan Tembaga SD 5/2, Bandar Sri Damansara, Kepong, 52200 Kuala Lumpur, Malaysia.
TEL	:	603-6273 4269/6274 8358
FAX	:	603-6273 4267
EMAIL	:	info@amoil.com.my

SECTION 2 : HAZARDS IDENTIFICATION

According to Regulation 2012 OSHA Hazard Communication Standard; 29 CFR Part 1910.1200

Classification of the product

Flam. Liq.	4	Flammable liquids
Skin Corr./Irrit.	2	Skin corrosion/irritation
Eye Dam./Irrit.	2A	Serious eye damage/eye irritation
Repr.	1B (unborn child)	Reproductive toxicity
STOT SE	3 (irritating to respiratory system)	Specific target organ toxicity — single exposure

Pictograms:

Signal Word: Danger

Hazard Statements:

H227:	Combustible.
H319:	Causes serious eye irritation
H315:	Cause skin irritation.
H335:	May cause respiratory irritation.
H360:	May damage the unborn child.

Precautionary Statement (Prevention):

P280: Wear protective glove / protective clothing / eye protection /face
P271: protection.
P210: Use only outdoor or in a well- ventilated area.
Keep away from heat, hot surfaces, sparks, open flames and other
P260: ignition sources. No smoking.
P202: Do not breathe dust / gas/ mist/ vapors.
P264: Do not handle until all safety precautions have been read and understood.
Wash with plenty of water and soap thoroughly after handling.

Precautionary Statements (Response):

P312: Call a POISON CENTER or doctor / physician if you fell unwell.
P305 + P351 + P338: IF IN EYES: Rinse cautiously with water of several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P304 +P340: If Inhaled: Remove person to fresh air and keep comfortable for breathing.
P303 +P352: IF ON SKIN: Wash with plenty of soap and water.
P308 +P311: IF exposed or concerned: Call a POISON CENTER or doctor / physician.
P332 +P313: If skin irritation occurs: Get medical advice / attention.
P337 +P311: If eye irritation persists: Call a POISON CENTER or doctor /physician.
P362 +P364: Take off contaminated clothing and wash it before reuse.
P370 +P378: In case of fire: Use water spray, dry powder, foam or carbon dioxide for extinction.

Precautionary Statements (Storage):

P233: Keep container tightly closed.
P403 +P235: Store in a well – ventilated place. Keep cool.
P405: Store locked up.

Precautionary Statements (Disposal):

P501: Dispose of contents / container to hazardous or special waste collection point.

SECTION 3 : COMPOSITION & INGREDIENTS

CAS Number	Weight%	Chemical Name
872-50-4	>=99.7 - <=99.9%	N- Methylpyrrolidone
60544-40-3	>=0.05 - <=0.4%	Pyrrolidinone, dimethyl

SECTION 4 : FIRST AID MEASURES

If inhaled:
Remove the affected individual into fresh air and keep the person calm. Assist in breathing if necessary. Immediate medical required.

If on skin:
Wash affected areas thoroughly with soap and water. If irritation develops, seek medical attention.

If in eyes:
In case of contact with eyes, rinse immediately for at least 15 minutes with plenty of water. Seek medical attention

If swallowed:
Rinse mouth and then drink plenty of water. Induce vomiting. Never induce vomiting or give anything by mount if the victim is unconscious or having convulsions. Immediate medical attention required.

Most important symptoms and effects, both acute and delayed

Symptoms:
The most important know symptoms and effects are described in the labelling (see section 2) and / or in section 11.

Indication of any immediate medical attention and special treatment needed

<u>Note to physician</u>
Treatment:
Treat according to symptoms (decontamination, vital functions), no know specific antidote.

SECTION 5 : FIRE FIGHTING MEASURES

Extinguishing media
Suitable extinguishing media:
Water spray, dry power, foam, carbon dioxide.

Special hazards arising from the substance or mixture
Hazards during fire- fighting:
Carbon monoxide, carbon dioxide, nitrous gases. Under certain conditions in case of fire other hazardous combustion products may be generated.

Advice for fire-fighters
Protective equipment for fire-fighting:
Firefighters should be equippend with self – contained breathing apparatus and

turn - out gear.

Further information:
Collect contaminated extinguishing water separately, do not allow to reach Sewage or effluent systems.

SECTION 6 : ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures
Wear appropriate respiratory protection. Use personal protective clothing.
Ensure adequate ventilation.

Environmental precautions
Do not discharge into the subsoil /soil. Do not discharge into drains / surface water/ groundwater. Retain and dispose of contaminated wash water. Dispose of in compliance with the environmental protection requirements.

Methods and material for containment and cleaning up
For small amounts: Pick up with absorbent material (e.g. sand, sawdust, general-purpose binder). Dispose of absorbed material in accordance with regulation.

SECTION 7 : HANDLING & STORAGE

Precautions for safe handling
Ensure thorough ventilation of store and work areas. Product should be worked up in closed equipment as far as possible.

Avoid contact with skin and eyes. Wear suitable gloves and eye / face protection.

Protection against fire and explosion:
The product is combustible.

Conditions for safe storage, including any incompatibilities
Further information on storage conditions: Containers should be stored tightly sealed in dry place.

SECTION 8 : EXPOSURE CONTROL / PERSONAL PROTECTION

No occupational exposure limits known.

Advice on system design:
Provide local exhaust ventilation to maintain recommended P.E.L

<u>Personal protective equipment</u>

Respiratory protection:
Wear a NIOSH – certified (or equivalent) organic vapour respirator. Observe OSHA regulations for respirator use (29 CFR 1910.134)

Hand Protection:
Wear chemical resistant protective gloves, butyl rubber (butyl) -0.7mm coating thickness, Consult with glove manufacturer for testing data.

Eye protection:
Safety glasses with side-shields. Wear face shield or tightly fitting safety goggles (chemical goggles) if splashing hazard exists.

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. head protection, apron, protective boost, chemical- protection suit.

General safety and hygiene measures:
Handle in accordance with good industrial hygiene safety practice. Avoid contact with skin the skin, eyes and clothing. Females in early pregnancy must never be exposed to the substance. Under no circumstances should the product come into contact with skin of pregnant women or be inhaled by them. Eye wash fountains and safety showers must be easily accessible. Wear protective clothing as necessary to minimize contact. Wash soiled clothing immediately. When using do not eat or drink. When using do not smoke. Gloves must be inspected regularly and prior to each use. Replace if necessary (e.g. pinhole leaks). Handle in accordance with good industrial hygiene and safety practice.

SECTION 9 : PHYSICAL & CHEMICAL PROPERTIES

Form	:	Liquid	
Odour	:	Mild, amine - like	
Odour threshold	:	Not determined due to potential health hazard by inhalation.	
Colour	:	Clear colourless	
pH value	:	8.5 - 10	
Melting point	:	-23.6 °C(760 mmHg)	
Boiling point	:	204.3 °C (760 mmHg)	
Flash point	:	196 °F	(ASTM D93)
Flammability	:	Combustible liquid	
Lower explosion limit	:	For liquids not relevant for classification and labelling. The lower explosion point may be 5-15°C below the flash point.	
Upper explosion limit	:	For liquids not relevant for classification and labelling.	
Autoignition	:	245 °C	
Vapour pressure	:	0.32 hPa (20°c)	(measured)

Density	:	1.028 g / cm3 (25°C)	(DIN 51757)
Relative density	:	1.0300 (20°C)	
Vapour density	:	Not determined	
Partitioning coefficient n-octanol / water (log Pow)	:	-0.46 (25°C)	(OECD Guideline 107)
Self – ignition temperature	:	Not self -igniting	
Thermal decomposition	:	365 °C, > 100kJ /kg (DSC(DIN51007)) Thermal decomposition above the indicated temperature is possible. It is not a self – decompositionable substance.	
Viscosity, dynamic	:	1.661 mPa.s (25°C)	
Solubility in water	:	Miscible, Literature data	
Solubility (qualitative)	:	Miscible solvents (s): organic solvents	
Molar mass	:	99.13 g /mol	
Evaporation rate	:	Value can be approximated from Henry's Law Constant or vapor Pressure.	

SECTION 10 : STABILITY & REACTIVITY

Reactivity:
No hazardous reactions if stored and handled as prescribed / indicated.

Corrosion to metals:
No corrosive effect on metal.

Oxidizing properties:
Not fire – propagating (other). Formation of Flammable gases:
Forms not flammable gases in the presence of water.

Chemical stability:
The product is stable if store and handled as prescribed /indicated.

Possibility of hazardous reaction:
Exothermic reaction. Reacts with strong acids and alkalies. Reacts with oxidizing agents

Conditions to avoid:
Avoid all sources of ignition: heat sparks open flame.

Incompatible materials:
Strong acids, oxidizing agents

Hazardous Decomposition Products

Decomposition products:
Hazardous decomposition products: carbon monoxide, carbon dioxide, nitrogen oxides.

Thermal decomposition:
365 °C (DSC (DIN 51007)). Thermal decomposition above the indicated temperature is possible. It is not a self decompositionable substance.

SECTION 11 : TOXICOLOGICAL INFORMATION

Primary routes of exposure
Routes of entry for solids and liquids are ingestion and inhalation, but may include eye or skin contact. Routes of entry for gases include inhalation and eye contact. Skin contact may be a route of entry for liquefied gases.

Acute toxicity /Effects

Acute toxity:
Assessment of acute toxicity: Of low toxicity single ingestion. Virtually nontoxic by inhalation. Virtually nontoxic after a single skin contact.

Oral:
Type of value: LD50
Species: rat (male / female)
Value: 4,150 mg/ kg (OECD Guideline 401)
Literature data.

Inhalation:
Type of value: LD50
Species: rat (male / female)
Value: > 5.1 mg / 1 (OEDC Guideline 403)
Exposure time: 4 h
An aerosol was tested.
Limit concentration test only (LIMIT test). No mortality was observed.

Dermal:
Type of value: LD50
Species: rat (male / female)
Value: > 5,000 mg/ kg (OECD Guideline 402)
Literature data.

Assessment other acute effects:
Assessment of STOT single: Causes temporary irritation of tract.

Irritation / corrosion:
Assess of irritation effects: Eye contact causes irritation. Skin contact causes irritation. Causes temporary irritation of the respiratory tract.

Skin:
Species: Rabbit
Result: Slightly irritating
Method: Draize test
Literature data

Eye:
Species: Rabbit
Result: Slightly irritation
Method: Draize test
Literature data.

Sensitization:
Assessment of sensitization: Skin sensitizing effects were not observed in animal studies.

Mouse Local Lymph Node Assay (LLNA)
Species: Mouse
Result: Non- sensitizing.
Method: OECD Guideline 429
The product has not been tested. The statement has been derived from substances / products of a similar structure or composition.

Aspiration Hazard
Not applicable

Chronic Toxicity / Effects

Repeated dose toxicity:
Assessment of repeated dose toxicity. After repeated exposure the prominent effect is local irritation. The substance may cause damage to the testes after repeated inhalation of high doses. Experimental / calculated data: rat by inhalation 2 weak 10 dose;
rat by inhalation 2 week 10 dose.

Genetic toxicity:
Assessment of mutagenicity: The substance was no mutagenic in bacteria. No mutagenic effect was found in various tests with mammalian cell culture and mammals.

Carcinogenicity:

Assessment of carcinogenicity: In long- term animal studies in which the substance was given by inhalation a carcinogenic effect was not observed. In long- term studies in rats in which the substance was given by feed a carcinogenicity effect was no observed. In long –term studies in rodents exposed to high doses, a tumorigenic effect was found; however, these results are thought to be due to a rodent – specific liver effect that is not relevant to humans. The whole of the information assessable provides no indication of carcinogenic effect.

Reproductive toxicity:
Assessment of reproductive toxicity: As shown in animal studies, the product may cause damage to the testes after repeated high exposures that cause other toxic effects.

Teratogenicity:
Assessment of teratogenicity: The substance caused malformations / developmental toxicity in laboratory animals.

Symptoms of Exposure
The most important know symptoms and effects are described in the labelling (see section 2)
And / or section 11.

SECTION 12 :	ECOLOGICAL INFORMATION

Toxicity

Aquatic toxicity:
Assessment of aquatic toxicity: There is a high probability that the product is not acutely harmful to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Toxicity to fish:
LC50 (96h) > 500mg /l, Salmo gairdneri, syn.O mykiss (static). The details of the toxic effect relate to the nominal concentration.

Aquatic Invertebrates:
EC50 (24 h) >1,000 mg /l, Daphnia magna (DIN 38412 part 11, static)

The details of the toxic effect relate to the nominal concentration.

Aquatic plants:
EC50 (72h) > 500 mg/l, Scenedesmus subspicatus (DIN 38412 Part 9). The details of the effect relate to the nominal concentration.

Chronic toxicity to fish:
Study scientifically not justified.

Chronic toxicity to aquatic invertebrates:
No observed effect concentration (21d) 1.5 mg /l, Daphnia magna (OECD Guideline 202, part 2 semistatic). The detail of the effect relate to the nominal concentration.

Assessment of terrestrial toxicity:
Study scientifically not justified.

Microorganisms /Effect on activated sludge

Toxicity to microorganisms:
DIN EN ISO 8192 AQUATIC; activated sludge, industrial / EC50 (0.5 h): > 600 mg /l. The details of the toxic effect to the nominal concentration

Persistence and degradability

Assessment biodegradation and elimination (H20)
Readily biodegradable (according to OECD criteria

Elimination information:
73% BOD of the ThOD (28d) (OECD 301C; ISO 9408; 92/69/EEC, C.4-F) (aerobic, Inoculum conforming to MITI requirements (OECD 301C)) Readily biodradable (according to OECD criteri).

Bioaccumulative potential

Assessment bioaccumulation potential
Because of the n- octanol / water distribution coefficient (log pow) accumulation in organisms is not to be expected.

Mobility in soil

Assessment transport between environmental compartments
The substance will rapidly evaporate into the atmosphere from water surface. Adsorption to solid soil phase is not expected.

Additional information

Sum parameter
Chemical oxygen demand (COD): (DIN 38409 Part 41) Approx 1,600 mg/ g
Biochemical oxygen demand (BOD) Incubation period 5 d: <2 mg /g
Adsorbable organically –bound halogen (AOX)

SECTION 13 : DISPOSAL CONSIDERATIONS

Water disposal of substance:
Dispose of in accordance with national, state and local regulation. Do not discharge into waterways or sewer systems without authorization.

Container dispose:
Dispose of in a licensed facility. Recommend crushing, puncturing or other means to prevent unauthorized use of used containers.

SECTION 14 : TRANSPORT INFORMATION

Land transport (USDOT):
Classified as combustible liquid in containers greater than 119 gallons.

Sea transport (IMDG):
Not classified as a dangerous good under transport regulations.

Air transport (IATA /ICAO):
Not classified as dangerous good under transport regulations.

SECTION 15 : REGULATORY INFORMATION

Federal Regulations

Registration status (Chemical):
TSCA, US released / listed

EPCRCRA 311.312 (Hazard categories

EPCRA 313:

CAS Number	Chemical Name
872-50-4	N-Methylpyttolidone

CERCLA RQ	CAS Number	Chemical Name
100 LBS	74-89-5	Methylamine

State Regulation	CAS Number	CAS Number
State RTK		
NJ	872-50-4	N- Methylpyrrolidone
PA	872-50-4	N- Methylpyrrolidone

WARNING:
This Product contains a chemical (S) known to the state of California to cause birth defects or other reproductive Harm.

NEPA Hazard codes:
Health: 2 Fire : 2 Reactivity: 0 Special:

HIMIS III rating

Health: 2®	Flammability : 2	Physical hazard: 0

Assessment of the hazard classes according to UN GHS criteria (most recent version)

Eye Dam. / Irrit	2A	Serious eye damage / eye irritation.
Skin Corr. / Irrit.	2	Skin corrosion / irritation
STOT SE	3 (irritating to respiratory system)	Specific target organ toxicity- single exposure
Repr	1B (unborn Child)	Reproductive toxicity
Flam. Liq.	4	Flammable liquids
Acute Tox.	5 oral	Acute toxicity

SECTION 16: OTHER INFORMATION DATA

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

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